

ASHIYA AB, Japan

WMO: 478030

Lat: 33.883N

Lon: 130.653E

Elev: 98

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.4	32.3	17.5	13.3	37.1	19.4	14.7	38.5	27.4	37.7	25.0	39.8	9.8	160	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.4	91.0	78.4	88.2	78.4	86.4	77.9	80.1	86.3	79.2	85.4	78.7	84.8	11.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.7	149.6	84.2	77.3	142.6	82.9	76.9	140.8	82.6	43.9	87.0	43.1	86.0	42.2	85.2	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.6	19.7	17.6	27.5	94.0	1.5	1.6	26.4	95.1	25.5	96.0	24.7	96.9	23.6	98.1		
			25.2	81.9	1.6	1.1	24.0	82.7	23.1	83.4	22.2	84.0	21.0	84.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.8	42.2	43.5	48.6	56.8	64.9	71.6	79.1	81.3	74.3	64.8	55.1	46.3
	DBStd	14.16	4.31	5.15	5.41	5.40	4.37	3.97	4.07	3.39	4.47	4.75	5.48	4.98
	HDD50	689	245	193	91	8	0	0	0	0	0	0	15	137
	HDD65	3065	707	602	507	251	56	1	0	0	0	61	299	581
	CDD50	4638	3	11	49	213	462	648	903	971	729	460	168	21
	CDD65	1537	0	0	0	6	52	199	438	506	279	55	2	0
	CDH74	12076	0	0	1	29	201	808	3942	5223	1720	149	3	0
	CDH80	3563	0	0	0	3	19	134	1228	1846	325	8	0	0
Wind	WSAvg	8.3	9.7	9.3	9.1	8.3	7.6	7.2	6.9	7.6	7.9	8.0	8.3	9.5
Precipitation	PrecAvg	69.9	3.0	3.3	4.4	5.2	6.0	11.1	12.5	7.4	7.0	3.3	3.7	2.9
	PrecMax	95.0	8.6	6.0	10.5	9.3	12.6	20.0	26.9	17.1	19.7	8.9	6.9	5.3
	PrecMin	41.0	1.2	1.3	2.3	1.4	1.4	1.6	2.4	1.0	2.7	0.6	1.1	0.8
	PrecStd	12.7	1.5	1.2	1.8	2.2	2.9	5.1	7.3	4.3	3.7	2.2	1.7	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.1	65.8	69.7	78.5	82.4	87.5	91.7	93.6	89.5	80.7	73.1	62.8
		MCWB	54.0	58.3	57.2	65.1	66.5	76.9	78.5	78.5	77.6	71.1	63.6	57.3
	2%	DB	55.1	58.9	64.5	71.9	78.8	83.9	89.7	91.6	86.1	77.3	69.4	59.3
		MCWB	49.6	53.5	54.9	60.4	65.2	73.8	78.4	78.5	77.1	68.8	62.0	52.9
	5%	DB	52.1	55.4	60.9	69.4	75.6	80.8	87.9	89.7	83.9	75.3	66.5	57.4
		MCWB	46.3	49.5	53.8	59.5	64.3	72.2	78.5	78.3	75.7	67.4	59.6	51.0
	10%	DB	50.2	52.2	57.6	66.2	73.3	78.7	86.1	87.9	81.0	73.3	64.5	55.4
		MCWB	44.6	46.7	51.5	58.0	63.6	71.7	78.0	78.4	74.2	65.7	58.3	49.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.7	59.1	60.7	68.2	72.8	78.1	80.7	81.6	79.5	74.2	66.5	59.4
		MCDB	58.1	64.1	64.4	74.1	77.1	84.3	87.2	87.6	85.1	78.4	68.8	61.3
	2%	WB	51.1	54.8	57.3	63.8	69.4	76.4	79.7	80.6	78.3	71.6	63.5	54.9
		MCDB	53.2	58.0	62.0	69.3	73.8	81.7	86.1	86.9	83.7	75.9	67.5	58.1
	5%	WB	47.7	51.0	55.0	61.5	67.3	74.8	79.0	79.8	77.0	69.3	61.5	52.5
		MCDB	51.2	53.9	59.9	66.8	72.3	78.8	85.3	86.0	81.8	73.3	65.5	56.2
	10%	WB	45.2	48.0	52.6	59.5	65.5	73.2	78.4	79.0	75.8	67.1	59.3	50.3
		MCDB	49.0	51.9	57.4	64.4	71.0	76.7	84.4	85.3	80.2	71.7	63.8	54.1
Mean Daily Temperature Range	5% DB	MDBR	11.8	12.8	13.8	14.6	13.9	11.0	10.4	11.4	11.4	13.7	13.9	12.7
		MCDBR	15.6	17.6	19.0	19.6	18.6	15.0	13.2	14.1	13.3	15.6	16.0	15.8
		MCWBR	11.5	13.1	12.6	11.2	9.3	7.4	5.6	5.3	6.2	8.4	10.2	11.4
	5% WB	MCDBR	13.9	15.0	15.9	14.8	13.6	11.0	11.7	12.2	11.2	11.3	13.0	13.2
		MCWBR	12.0	13.0	12.3	9.9	9.2	6.8	5.6	5.4	6.0	7.5	9.7	11.3
Clear-Sky Solar Irradiance	taub	0.430	0.471	0.533	0.554	0.560	0.575	0.570	0.533	0.503	0.458	0.431	0.422	
	taud	2.058	1.985	1.838	1.842	1.889	1.937	1.997	2.100	2.134	2.156	2.142	2.093	
	Ebn at Noon	236	240	236	239	239	235	235	241	243	243	235	230	
	Edh at Noon	42	50	63	65	63	60	56	50	46	42	38	38	
All-Sky Solar Radiation	RadAvg	684	913	1261	1530	1670	1440	1556	1631	1298	1099	813	637	
	RadStd	62	91	90	128	192	160	237	182	130	114	77	43	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page